ModelIt!

Student Activity Pack

# CRISPR Gene Drives: Rewriting Evolution

Modeling How Engineered Genes Spread Through Wild Populations

Grade 09L2 | NGSS HS-LS3-1, HS-LS4-5 | Cause and Effect

STUDENT NAME _____ DATE _____



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# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. What is CRISPR and how does it edit genes?

2. If you could eliminate all mosquitoes to stop malaria, should you? What might go wrong?

3. Draw a Punnett square showing normal 50-50 inheritance. Now imagine one allele gets passed 99 percent of the time.

4. What does it mean to edit an entire wild species?

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| Identifying Components | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| Establishing Relationships | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| Running Simulations | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| Explaining Results | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

Gene Drive Frequency | Wild-Type Allele Frequency | Population Size | Fitness Cost | Resistance Evolution | Inheritance Bias | Ecological Impact

| EXTERNAL | INTERNAL |
|---|---|
| | |

### ? Explain your reasoning: Why did you sort each component the way you did?

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? **Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: Ideal Drive

99 percent inheritance, no fitness cost

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: Realistic Drive

95 percent inheritance, moderate fitness cost

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: Resistance Challenge

95 percent inheritance, moderate cost, high resistance rate

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Daisy Chain Mechanism** — A self-limiting gene drive design where the drive loses power after a set number of generations

**Off-Target Species Effects** — The probability of the gene drive accidentally transferring to related species through hybridization

**Community Consent Level** — The degree of informed consent from local communities where the gene drive would be released

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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### The Big Question

**Should we have the power to edit an entire species -- and what happens if we get it wrong?** Use evidence from your simulation to support your answers.

1. How does your model show the race between gene drive spread and resistance evolution?

2. Why is ecological impact the hardest component to predict in your model?

3. What ethical considerations arise when one country releases a gene drive that could cross borders?

4. How would you decide whether the benefit of eliminating malaria outweighs the ecological risk?

5. What safety mechanisms could make gene drives reversible?

# 8 STEM Challenge: Design a Gene Drive Safety Protocol

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## YOUR ENGINEERING MISSION

Design a gene drive deployment strategy for malaria mosquito suppression that maximizes disease reduction while minimizing ecological risk.

### THE SCENARIO

Malaria kills 600,000 people per year, mostly children in sub-Saharan Africa. A gene drive targeting Anopheles mosquitoes could potentially eliminate malaria. But the ecological consequences of removing a species are unknown. Design a phased deployment with safety mechanisms.

## Engineering Design Thinking

1. WOULD YOU USE A SUPPRESSION DRIVE (REDUCE POPULATION) OR MODIFICATION DRIVE (MAKE MOSQUITOES UNABLE TO CARRY MALARIA)?

2. HOW WOULD YOU CONTAIN THE DRIVE TO A SPECIFIC GEOGRAPHIC REGION?

3. WHAT MONITORING SYSTEMS WOULD DETECT ECOLOGICAL SIDE EFFECTS?

4. HOW WOULD YOU OBTAIN INFORMED CONSENT FROM COMMUNITIES WHERE THE DRIVE IS RELEASED?

## My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

## Real Career Connection

Genetic Engineers and Bioethicists work at the intersection of technology and society, designing safe genetic modifications and evaluating their ethical implications, earning $70,000-$150,000/year at biotech companies, universities, and government agencies.